

Literature-implied full answer for arXiv:math/0702269

Balanced monomials violate the Schur–Agler ball estimate

Status. literature_implied_answer (full negative answer). The counterexample below completely answers the existence question in Open Problem 2 of Anderson–Dritschel–Rovnyak. Its ingredients are already implicit in the sharp coefficient theorem of Dai–Chen–Pan, but the supporting paper does not cite the open problem or point out the contradiction with the source estimate. Accordingly, this packet claims the identification, not novelty for the monomial extremal itself.

Source question and estimate. On page 18, in Section 5 under “Some open problems”, Anderson, Dritschel, and Rovnyak ask whether there are Schur functions outside the Schur–Agler class on the polydisk or unit ball which fail the estimates proved in Sections 3 and 4. For the Euclidean ball $\mathbb{B}_d$, their Theorem 12, estimate (42), says that every ball Schur–Agler function φ satisfies

$$|\partial_1^{n_1} \cdots \partial_d^{n_d} \varphi(z)| \leq d^{(n-1)/2} n_1! \cdots n_d! \frac{1 - |\varphi(z)|^2}{(1 - \|z\|_2^2)(1 - \|z\|_2)^{n-1}}, \quad n = \sum_j n_j. \quad (1)$$

Supporting literature. Dai–Chen–Pan, Theorem 2, proves that if $f(z) = \sum_\alpha a_\alpha z^\alpha$ maps $\mathbb{B}_d$ to $\mathbb{D}$, then at the origin

$$|a_\alpha| \leq \sqrt{\frac{|\alpha|!}{\alpha^\alpha}} (1 - |a_0|^2),$$

and describes the monomial extremals. For the balanced multi-index $\alpha = (1, \dots, 1)$, their sharp factor is $d^{d/2}$, while (1) permits only $d^{(d-1)/2}$.

Theorem. *For every integer $d \geq 2$, the polynomial*

$$F_d(z) = d^{d/2} z_1 z_2 \cdots z_d$$

belongs to the Schur class on $\mathbb{B}_d$, does not belong to the ball Schur–Agler class, and violates estimate (1) at the origin by the exact factor $\sqrt{d}$. Hence Open Problem 2 has a full affirmative answer to its existence question.

Proof. For $z \in \mathbb{B}_d$, arithmetic–geometric mean gives

$$|F_d(z)| = d^{d/2} \prod_{j=1}^d |z_j| \leq d^{d/2} \left(\frac{\sum_j |z_j|^2}{d} \right)^{d/2} = \|z\|_2^d < 1.$$

Thus $F_d : \mathbb{B}_d \rightarrow \mathbb{D}$. Its completely mixed derivative is constant:

$$\partial_1 \partial_2 \cdots \partial_d F_d(0) = d^{d/2}.$$

In (1), take $n = d$, $n_1 = \cdots = n_d = 1$, and $z = 0$. Since $F_d(0) = 0$, the proposed right side is exactly $d^{(d-1)/2}$. The quotient of the two sides is $\sqrt{d} > 1$. Therefore F_d violates the estimate. Theorem 12 proves the estimate for every ball Schur–Agler function, so contraposition also shows that F_d is not Schur–Agler. $\square$

Smallest instance. On $\mathbb{B}_2$, $F_2(z_1, z_2) = 2z_1z_2$ is Schur because $2|z_1z_2| \leq |z_1|^2 + |z_2|^2 < 1$. At zero, $|\partial_1\partial_2F_2| = 2$, but (42) gives only $\sqrt{2}$.

The two examples singled out in the source

The counterexample above settles existence, but it is different from the two families the source explicitly asks about. The following elementary companion calculations settle both of those tests positively.

Proposition. *Every homogeneous quadratic Schur polynomial on $\mathbb{D}^d$ satisfies all the polydisk estimates of Anderson–Dritschel–Rovnyak. In particular, the Kaijser–Varopoulos polynomial satisfies them all.*

Proof. Let $p : \mathbb{D}^d \rightarrow \mathbb{D}$ be homogeneous of degree two and put $r = \|z\|_\infty$. Homogeneity gives $|p(z)| \leq r^2$. The source’s first-derivative estimate is automatic from the one-variable Schwarz–Pick lemma on the coordinate slice, because

$$|\partial_j p(z)| \leq \frac{1 - |p(z)|^2}{1 - |z_j|^2} \leq \frac{1 - |p(z)|^2}{\sqrt{1 - |z_j|^2} \sqrt{1 - r^2}}.$$

All derivatives of order at least three vanish. If $p = \sum_{|\alpha|=2} a_\alpha z^\alpha$, a torus Fourier integral gives $|a_\alpha| \leq 1$. Hence $|\partial_j^2 p| \leq 2$, while $|\partial_j \partial_k p| \leq 1$ for $j \neq k$.

For a second derivative, the right side of the source’s permutation-sum estimate is at least

$$2 \frac{1 - r^4}{1 - r} \geq 2.$$

The right side of its multi-index estimate is at least

$$\alpha! \frac{1 - r^4}{(1 - r^2)(1 - r)} = \alpha! \frac{1 + r^2}{1 - r} \geq \alpha!.$$

These dominate the pure and mixed second derivatives, respectively. □

Proposition. *For every Schur function $f : \mathbb{B}_d \rightarrow \mathbb{D}$, every pure-coordinate derivative $\partial_j^n f$ satisfies both ball estimates in the source. Consequently, every Alpay–Kaptanoğlu polynomial singled out in Open Problem 2 satisfies all of those estimates.*

Proof. Fix $z \in \mathbb{B}_d$ and a coordinate j . Set

$$a = \|\hat{z}_j\|_2, \quad R = \sqrt{1 - a^2}, \quad b = |z_j|, \quad r = \|z\|_2.$$

The slice $g(\lambda) = f(\hat{z}_j + R\lambda e_j)$ maps $\mathbb{D}$ into $\mathbb{D}$. Applying the sharp one-variable higher Schwarz–Pick estimate at $\lambda = z_j/R$ and rescaling gives

$$|\partial_j^n f(z)| \leq \frac{n! R (1 - |f(z)|^2)}{(1 - r^2)(R - b)^{n-1}}. \tag{2}$$

But

$$R - b = \frac{1 - r^2}{R + b} \geq \frac{1 - r^2}{1 + r} = 1 - r.$$

Thus (2) is at least as strong as the source’s Theorem 10 bound for a pure derivative, whose numerator contains $n!R$, and also as strong as its Theorem 12 bound, whose numerator contains $d^{(n-1)/2}n! \geq n!R$. The first-order case is the standard infinitesimal Schwarz–Pick inequality for the ball.

An Alpay–Kaptanoğlu polynomial has the form $z_1 + \sum_{k=1}^m c_k z_2^{2k}$. Every derivative of order at least two is either zero or pure in z_2 , while the first derivatives satisfy the universal first-order estimate. Hence all source estimates hold. $\square$

Verification and provenance. The Schur bound and derivative comparison for F_d are exact and use only AM–GM. The included checker confirms the exponent arithmetic for $2 \leq d \leq 20$; it is not used as proof. The run first searched its cheap registry, solution, and attempt indexes for the exact arXiv id and the phrases “Kaijser–Varopoulos”, “Schur–Agler”, and “Schwarz–Pick”. Bounded web and arXiv searches used the exact expressions “ $2z_1z_2$ Schur–Agler”, “Kaijser–Varopoulos Schwarz–Pick derivative”, and the source title. No paper explicitly stating an answer to Open Problem 2 was found. The closest decisive source was arXiv:1109.2791, whose Theorem 2 supplies the general sharp coefficient extremal but does not cite or identify the 2007 question. This is why the durable status is literature-implied rather than a new counterexample.

Scope. The existence question is completely answered on the unit ball in every dimension $d \geq 2$. The packet does not produce a violating function on a polydisk, settle sharpness of all source constants, or carry out the source’s Cartan-domain program. The companion propositions settle only the two explicit test families named in Open Problem 2.

Human-review recommendation. Verify the substitution $n = d$, $n_j = 1$, $z = 0$ in source estimate (42), and confirm the provenance classification against Dai–Chen–Pan Theorem 2. The mathematical counterexample itself is a one-line exact check.

References

- [1] J. M. Anderson, M. A. Dritschel, and J. Rovnyak, *Schwarz–Pick inequalities for the Schur–Agler class on the polydisk and unit ball*, Comput. Methods Funct. Theory 8 (2008), 339–361; arXiv:math/0702269.
- [2] S. Dai, H. Chen, and Y. Pan, *The Schwarz–Pick lemma of high order in several variables*, arXiv:1109.2791.
- [3] W. Rudin, *Function Theory in the Unit Ball of $\mathbb{C}^n$* , Springer, 1980.